

Introduction

Before You Begin

Thank you for your interest in this academic research study. This study examines how people evaluate passwords and interpret password-strength feedback. It typically takes **15–20 minutes** to complete.

What you will do

You will evaluate passwords created specifically for this study, view feedback from website Password Strength Meters (PSMs), and answer multiple-choice and free-text questions about your ratings, judgments, and decisions.

Do not enter or disclose any password that you use in real life. All passwords shown in this study are provided by the researchers.

Compensation

The study pays **\$3.00 through Prolific**. Depending on the study version and response quality, additional bonuses of up to **\$2.00** may be available.

Participation and response quality

Please complete the study independently and do not use AI tools or outside assistance. Responses may be reviewed for attention, completeness, and overall response quality. Clearly incomplete, nonsensical, or insufficiently engaged responses may be rejected.

Voluntary participation and data use

Your participation is voluntary. You may stop participating at any time by closing the survey and returning the study on Prolific. Study data will be securely stored and used only for academic research. The study has been reviewed by an Institutional Review Board (IRB).

Do you agree to participate in this study?

- ☐ I agree to participate in this study.
- ☐ I do not agree to participate in this study.

Survey Instrument

This document reproduces the questions shown in the study. Text in square brackets indicates values inserted dynamically during the survey. Conditional sections are labeled by study condition; individual participants saw only the questions applicable to their assigned condition.

In this study, **Below Strong** means any strength rating lower than **Strong**, such as Weak, Fair, or Medium.

Before continuing, please answer the following question.

In this study, what does “**Below Strong**” mean?

- ☐ Only a Weak rating
- ☐ I am not sure
- ☐ Any rating lower than Strong, such as Weak, Fair, or Medium
- ☐ The password had been used on another website
- ☐ The website rejected the password

Your previous answer was not correct.

Please note that in this study, **Below Strong** means any strength rating lower than **Strong**, such as Weak, Fair, or Medium.

It refers to the password's **strength rating**.

Please keep this definition in mind for the rest of the study.

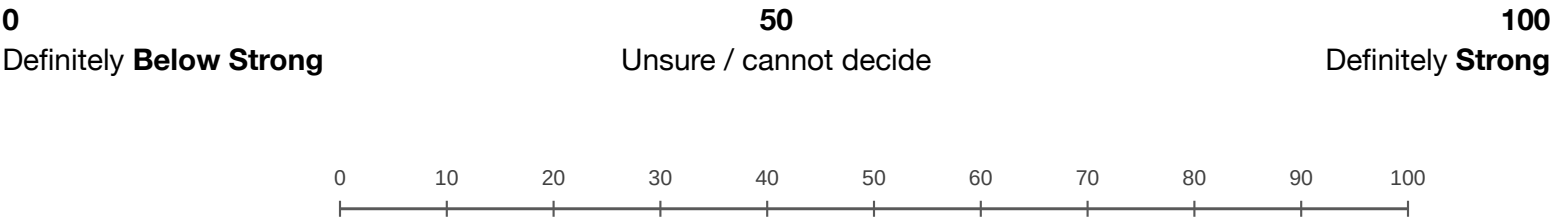
Baseline+PSMs

Imagine that you are choosing a password for a new online account.

Consider the password shown below based only on your own judgment.

[STUDY PASSWORD]

At this point, how strongly do you believe this password should be rated **Strong**?

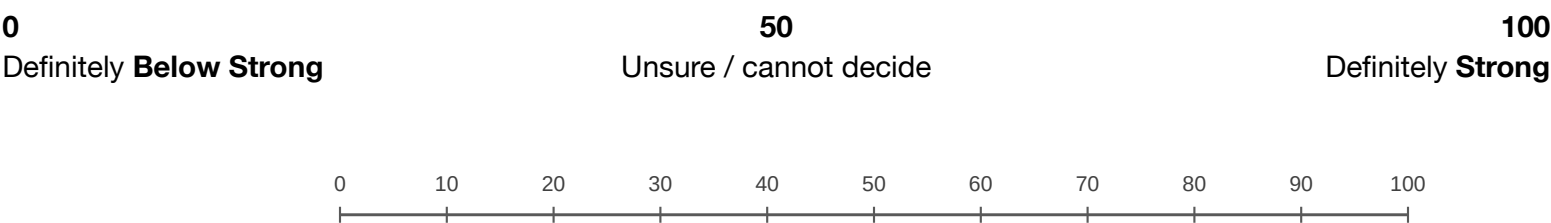


Suppose you are on **Website A** and check **this password** using its Password Strength Meter. It gives the following rating:

[STUDY PASSWORD]

Website A's Password Strength Meter
[STRONG OR BELOW STRONG]

Based on everything you have seen so far, how strongly do you believe this password should be rated **Strong**?

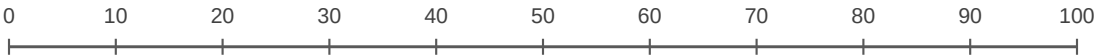
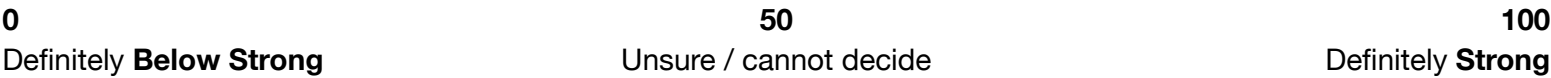


Later, you visit **Website B** and check **the same password** using its Password Strength Meter. It gives the following rating:

[STUDY PASSWORD]

Website B's Password Strength Meter
[STRONG OR BELOW STRONG]

Based on everything you have seen so far, how strongly do you believe this password should be rated **Strong**?



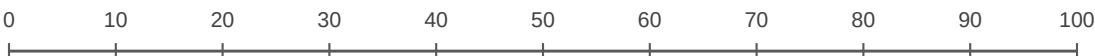
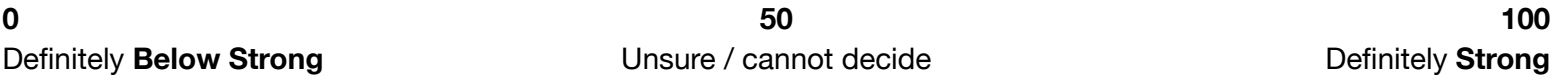
On another website, **Website C**, you check **the same password** again using its Password Strength Meter. It gives the following rating:

[STUDY PASSWORD]

Website C’s Password Strength Meter

[STRONG OR BELOW STRONG]

Based on everything you have seen so far, how strongly do you believe this password should be rated **Strong**?



You also check **the same password** using the Password Strength Meter on **Website D**. It gives the following rating:

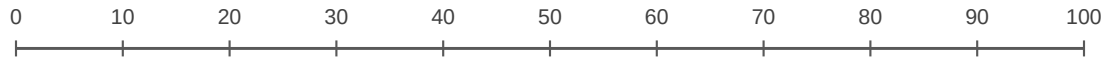
[STUDY PASSWORD]

Website D’s Password Strength Meter

[STRONG OR BELOW STRONG]

Based on everything you have seen so far, how strongly do you believe this password should be rated **Strong**?

0 **50** **100**
Definitely **Below Strong** Unsure / cannot decide Definitely **Strong**



Next Action

[STUDY PASSWORD]

After seeing these PSM ratings, imagine that you are now **deciding what to do with this password for a new online account**. Considering what you now think about **the password's strength**, what would you most likely do?

- ☐ Use the password as shown
- ☐ Modify the password before using it
- ☐ Choose a different password
- ☐ Seek more information before deciding

What most influenced your final judgment about the **password's strength**?

What was the main reason for **what you decided to do with the password**?

What information would you look for before deciding?

Thinking about the PSM ratings shown on Websites A–D, how much did they **influence what you decided to do with this password?**

- ☐ Not at all
- ☐ A little
- ☐ Moderately
- ☐ A lot
- ☐ A great deal

Thinking about how you judged the password's strength, from your initial judgment through Websites A–D, **what were the main reasons** your judgment **changed or stayed the same** as you went through the task?

Thinking about the PSM ratings shown on Websites A–D, how much did they **influence your judgment of the password's strength?**

- ☐ Not at all
- ☐ A little
- ☐ Moderately
- ☐ A lot
- ☐ A great deal

Conflict-condition questions (3:1 / 2:2 / 1:3)

Conflict-Condition Questions

[Shown only in the 3:1, 2:2, and 1:3 conditions.]

At any point, did you feel that a website's PSM rating **differed from** your own judgment of the password's strength?

- ☐ Yes
- ☐ No
- ☐ Not sure

When **your judgment of the password's strength** and a **website's PSM rating** differed, which had **more influence on your strength rating**?

- ☐ My own judgment had much more influence
- ☐ My own judgment had somewhat more influence
- ☐ They had about equal influence
- ☐ The PSM rating had somewhat more influence
- ☐ The PSM rating had much more influence
- ☐ It depended on the situation
- ☐ Not sure

What factors influenced whether you relied more on **your own judgment** or the **PSM rating**?

Did you notice that the PSMs on different websites **gave different ratings** for the same password?

- ☐ Yes
- ☐ No
- ☐ Not sure

When the PSMs on different websites gave **different ratings**, what did you rely on most when **judging the password's strength**?

- ☐ My own judgment of the password's strength
- ☐ The rating given by most of the PSMs
- ☐ The Below Strong rating
- ☐ The Strong rating
- ☐ A particular website's PSM
- ☐ All of the PSM ratings considered together
- ☐ It depended on the situation
- ☐ Other (please specify)

What factors influenced **what you relied on most** when judging the password's strength after seeing **different PSM ratings across websites**?

How difficult, if at all, did you find each of the following during the study?

	Not at all difficult	Slightly difficult	Moderately difficult	Very difficult	Extremely difficult	I did not experience this situation
Deciding on a strength rating when your judgment of the password's strength and a website's PSM rating differed .	☐	☐	☐	☐	☐	☐
Deciding on a strength rating when PSMs on different websites gave different ratings for the same password.	☐	☐	☐	☐	☐	☐
Deciding what to do with the password after seeing all four ratings.	☐	☐	☐	☐	☐	☐

At this point, how much do you **generally trust website Password Strength Meter ratings** when judging **password strength**?

- ☐ Not at all
- ☐ Slightly
- ☐ Moderately
- ☐ Very much
- ☐ Completely

After completing this task, how did your **overall trust** in website Password Strength Meters **change**, if at all?

- ☐ Decreased a great deal
- ☐ Decreased somewhat
- ☐ Did not change
- ☐ Increased somewhat
- ☐ Increased a great deal

What factors influenced the **change, or lack of change, in your trust** in website Password Strength Meter ratings?

Suppose you encountered **conflicting PSM ratings for the same password** in the future. What would you do **to deal with the conflicting ratings** before deciding **what to do with the password**?

Based on your experience in this task, imagine that you are choosing a password while signing up for a website. What would you **most likely do** in each situation?

	Use the password as planned	Modify the password before using it	Choose a different password	Seek more information before deciding	Not sure
Situation A: You would rate the password Strong , but the website's PSM rates it Below Strong .	☐	☐	☐	☐	☐

Use the
password as
planned

Modify the
password
before using
it

Choose a
different
password

Seek more
information
before
deciding

Not sure

Situation B: You would
rate the password **Below
Strong**, but the website's
PSM rates it **Strong**.



What factors most influenced **your choice in each situation**?

Situation A

You would rate the password **Strong**, but the website's PSM rates it **Below Strong**.

Situation B

You would rate the password **Below Strong**, but the website's PSM rates it **Strong**.

Scenario A: Strong → Below Strong

Suppose you already use a password for an account on one website. When you created that account, the website's PSM rated the password **Strong**.

Now, while creating **a new account on another website**, you check the same password using that website's PSM. It rates the password **Below Strong**.

For the new account, what would you most likely do?

- ☐ Use the same password
- ☐ Modify the password before using it
- ☐ Choose a different password
- ☐ Seek more information before deciding
- ☐ Not sure

For your existing account on the earlier website, what would you most likely do?

- ☐ Keep the existing password unchanged
- ☐ Modify the existing password
- ☐ Replace it with a different password
- ☐ Seek more information before deciding
- ☐ Not sure

Scenario A: Strong → Below Strong

Suppose you already use a password for an account on one website. When you created that account, the website’s PSM rated the password **Strong**.

Now, while creating a new account on another website, you check the same password using that website’s PSM. It rates the password **Below Strong**.

What factors most influenced **your choice for each account**?

New account

Existing account



Scenario B: Below Strong → Strong

Suppose you already use a password for an account on one website. When you created that account, the website's PSM rated the password **Below Strong**.

Now, while creating **a new account on another website**, you check the same password using that website's PSM. It rates the password **Strong**.

For the new account, what would you most likely do?

- ☐ Use the same password
- ☐ Modify the password before using it
- ☐ Choose a different password
- ☐ Seek more information before deciding
- ☐ Not sure

For your existing account on the earlier website, what would you most likely do?

- ☐ Keep the existing password unchanged
- ☐ Modify the existing password
- ☐ Replace it with a different password
- ☐ Seek more information before deciding
- ☐ Not sure

Scenario B: Below Strong → Strong

How realistic did each of the following situations seem compared with what you might encounter in real life?

	Not at all realistic	Slightly realistic	Moderately realistic	Very realistic	Extremely realistic
A website’s PSM rating differed from your own judgment of the password’s strength.	☐	☐	☐	☐	☐
PSMs on different websites gave different ratings for the same password.	☐	☐	☐	☐	☐

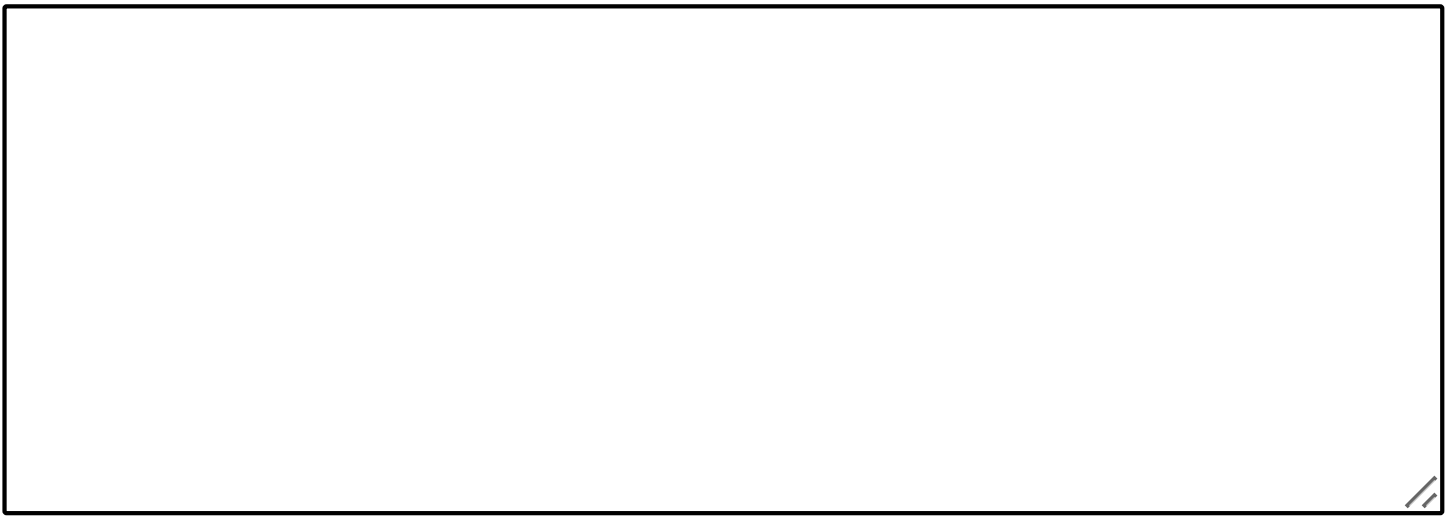
When judging the password’s strength, did you treat any of the PSMs on Websites A–D as **more or less reliable than the others**?

- ☐ Yes
- ☐ No
- ☐ Not sure

Compared with the other PSMs, how reliable did you consider each of the following?

	Much less reliable	Somewhat less reliable	About as reliable	Somewhat more reliable	Much more reliable
PSM on Website A	☐	☐	☐	☐	☐
PSM on Website B	☐	☐	☐	☐	☐
PSM on Website C	☐	☐	☐	☐	☐
PSM on Website D	☐	☐	☐	☐	☐

What were the main reasons for your **reliability judgments** about the PSMs on Websites A–D?



Unanimous-feedback questions (4:0 / 0:4)

Unanimous-Feedback Questions

[Shown only in the 4:0 and 0:4 conditions.]

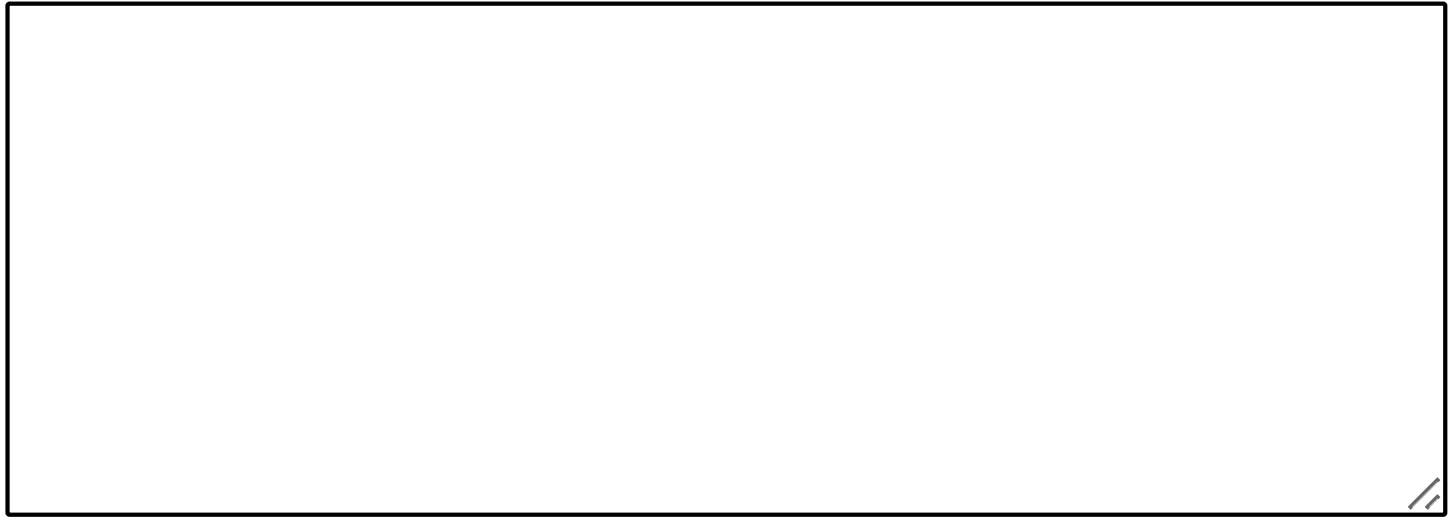
At any point, did you feel that a website's PSM rating **differed from** your own judgment of the password's strength?

- ☐ Yes
- ☐ No
- ☐ Not sure

When **your judgment of the password's strength** and a **website's PSM rating** differed, which had **more influence on your strength rating**?

- ☐ My own judgment had much more influence
- ☐ My own judgment had somewhat more influence
- ☐ They had about equal influence
- ☐ The PSM rating had somewhat more influence
- ☐ The PSM rating had much more influence
- ☐ It depended on the situation
- ☐ Not sure

What factors influenced whether you relied more on **your own judgment** or the **PSM rating**?



Did you notice that the PSMs on **all four websites gave the same rating** for the same password?

- ☐ Yes
- ☐ No
- ☐ Not sure

How much, if at all, did seeing the **same PSM rating repeatedly** influence your **judgment of the password's strength**?

- ☐ Not at all
- ☐ A little
- ☐ Moderately
- ☐ A lot
- ☐ A great deal

What factors influenced **how much the repeated PSM ratings affected** your judgment of the password's strength?

How difficult, if at all, did you find each of the following during the study?

	Not at all difficult	Slightly difficult	Moderately difficult	Very difficult	Extremely difficult	I did not experience this situation
Deciding on a strength rating when your judgment of the password's strength and a website's PSM rating differed	☐	☐	☐	☐	☐	☐
Deciding what to do with the password after seeing all four ratings.	☐	☐	☐	☐	☐	☐

At this point, how much do you **generally trust website Password Strength Meter ratings** when judging **password strength**?

- ☐ Not at all
- ☐ Slightly
- ☐ Moderately
- ☐ Very much
- ☐ Completely

After completing this task, how did your **overall trust** in website Password Strength Meters **change**, if at all?

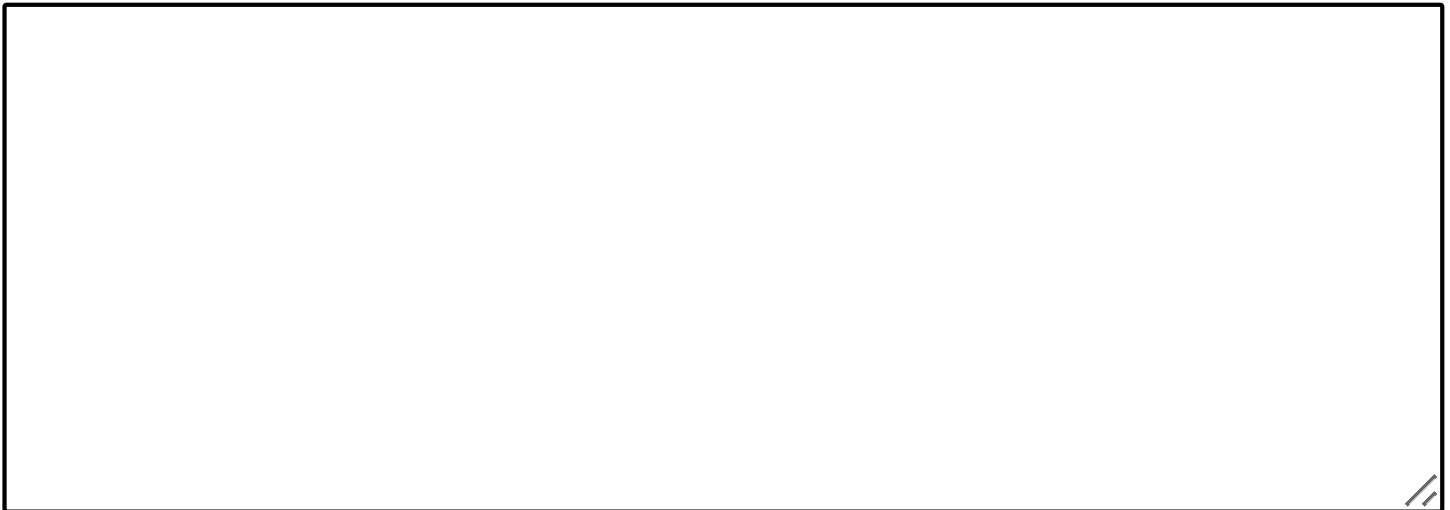
- ☐ Decreased a great deal

- ☐ Decreased somewhat
- ☐ Did not change
- ☐ Increased somewhat
- ☐ Increased a great deal

What factors influenced the **change, or lack of change, in your trust** in website Password Strength Meter ratings?



Suppose you encountered **conflicting PSM ratings for the same password** in the future. What would you do **to deal with the conflicting ratings** before deciding **what to do with the password**?



Based on your experience in this task, imagine that you are choosing a password while signing up for a website. What would you **most likely do** in each situation?

	Use the password as planned	Modify the password before using it	Choose a different password	Seek more information before deciding	Not sure
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Situation A: You would rate the password **Strong**, but the website’s PSM rates it **Below Strong**.

☐	☐	☐	☐	☐
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Situation B: You would rate the password **Below Strong**, but the website’s PSM rates it **Strong**.

☐	☐	☐	☐	☐
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What factors most influenced **your choice in each situation**?

Situation A

You would rate the password **Strong**, but the website’s PSM rates it **Below Strong**.

Situation B

You would rate the password **Below Strong**, but the website’s PSM rates it **Strong**.

Before this study, how often had you experienced each of the following?

Never Rarely Sometimes Often Very often Not sure

A website's PSM rating **differed from** your own judgment of the password's strength.

☐ ☐ ☐ ☐ ☐ ☐

PSMs on different websites gave **different ratings** for the same password.

☐ ☐ ☐ ☐ ☐ ☐

How realistic did each of the following situations seem compared with what you might encounter in real life?

Not at all realistic Slightly realistic Moderately realistic Very realistic Extremely realistic

A website's PSM rating **differed from** your own judgment of the password's strength.

☐ ☐ ☐ ☐ ☐

PSMs on different websites gave the **same rating** for the same password.

☐ ☐ ☐ ☐ ☐

When judging the password's strength, did you treat any of the PSMs on Websites A–D as **more or less reliable than the others**?

- ☐ Yes
☐ No
☐ Not sure

Compared with the other PSMs, how reliable did you consider each of the following?

Much less reliable Somewhat less reliable About as reliable Somewhat more reliable Much more reliable

PSM on Website A

☐ ☐ ☐ ☐ ☐

PSM on Website B

☐ ☐ ☐ ☐ ☐

PSM on Website C

☐ ☐ ☐ ☐ ☐

Much less
reliable

Somewhat
less reliable

About as
reliable

Somewhat
more reliable

Much more
reliable

PSM on Website D

☐☐☐☐☐

What were the main reasons for your **reliability judgments** about the PSMs on Websites A–D?

attention

Which statement best describes what a Password Strength Meter does?

- ☐ It estimates and displays how strong a password is.
- ☐ It stores the password so the user can log in later.
- ☐ It decides whether the user is allowed to create an account.
- ☐ I am not sure.

Post-task background-for both

Before this study, how much did you generally **rely on** website Password Strength Meter ratings when choosing passwords?

- ☐ Not at all
- ☐ Slightly
- ☐ Moderately

- ☐ Very much
- ☐ Completely
- ☐ Not sure

Do you currently use a password manager to store or generate passwords?

- ☐ Yes, regularly
- ☐ Yes, occasionally
- ☐ No
- ☐ I am not sure what a password manager is

Before this study, **how often did you use the same password** for more than one online account?

- ☐ Never
- ☐ Rarely
- ☐ Sometimes
- ☐ Often
- ☐ Very often
- ☐ Prefer not to say

How would you rate your level of knowledge about password security?

- ☐ Not at all knowledgeable
- ☐ Slightly knowledgeable
- ☐ Moderately knowledgeable
- ☐ Very knowledgeable
- ☐ Extremely knowledgeable

Do you have education, training, or work experience in computer science, information technology, cybersecurity, or a related field?

- ☐ Yes
- ☐ No
- ☐ Prefer not to say

Demographics

How old are you?

- ☐ 18-24 years old
- ☐ 25-34 years old
- ☐ 35-44 years old
- ☐ 45-54 years old
- ☐ 55-64 years old
- ☐ 65+ years old
- ☐ Prefer not to say

How do you describe your gender?

- ☐ Male
- ☐ Female
- ☐ Non-binary / third gender
- ☐ Prefer to self-describe
- ☐ Prefer not to say

What is the highest level of education you have completed?

- ☐ Some high school or less
- ☐ High school diploma or GED
- ☐ Some college, but no degree
- ☐ Associate or technical degree
- ☐ Bachelor's degree
- ☐ Graduate or professional degree (MA, MS, MBA, PhD, JD, MD, DDS etc.)
- ☐ Prefer not to say

Which race or races do you identify with?

- ☐ White or Caucasian

- ☐ Black or African American
- ☐ American Indian or Alaska Native
- ☐ Asian
- ☐ Native Hawaiian or Other Pacific Islander
- ☐ Prefer not to say
- ☐ Other

Which category best describes your primary field of study or occupation?

- ☐ Arts and Humanities
- ☐ Skilled Trades
- ☐ Business and Management
- ☐ Health and Medicine
- ☐ Social Sciences
- ☐ Computer Science and Information Technology
- ☐ Natural Sciences
- ☐ Education
- ☐ Engineering and Technology
- ☐ Prefer not to say
- ☐ Other

The end

Debriefing

The password-strength ratings shown during this study were part of the experimental design and were presented for research purposes.

Please do not use any password shown in this study for a real account.

Thank You

Thank you for completing this study. We appreciate your contribution to our research.